



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 10 • STANDARD MPP.7

Math is Boundless

Lesson 10-5 • Teacher Copy

Name: _____

Date: _____

Class: _____



TODAY'S OBJECTIVES

Content Objective: I can create and describe geometric designs using repetition, patterns, and rhythm, and use a pattern unit to predict what comes next.

Language Objective: I can describe a design using the words repetition, pattern, pattern unit, and rhythm.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

repetition

pattern

pattern unit

rhythm

predictability



Tap any word to see what it means and a picture.

1

— The same object, shape, or form repeated multiple times.

2

— A set of objects or shapes that are repeated and arranged in a regular order.

3

— The smallest chunk of a pattern that repeats.

4

— A design element created by varying the placement, size, orientation, or order of the same objects or shapes.

5

— Being able to know what comes next because the design follows a rule.

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

COMPARE

How is a design with rhythm different from a pattern?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- ☐ **repetition** — The same object, shape, or form repeated multiple times.
repetición — El mismo objeto, figura o forma repetido varias veces.

- ☐ **pattern** — A set of objects or shapes that are repeated and arranged in a regular order.
patrón — Un conjunto de objetos o figuras que se repiten y se organizan en un orden regular.

- ☐ **pattern unit** — The smallest chunk of a pattern that repeats.
unidad del patrón — La parte más pequeña de un patrón que se repite.

- ☐ **rhythm** — A design element created by varying the placement, size, orientation, or order of the same objects or shapes.
ritmo — Un elemento de diseño creado al variar la posición, el tamaño, la orientación o el orden de los mismos objetos o figuras.

- ☐ **predictability** — Being able to know what comes next because the design follows a rule.
previsibilidad — Poder saber qué viene después porque el diseño sigue una regla.

Say it first: Say your idea to a partner or quietly to yourself before you write.

Di tu idea a un compañero o en voz baja antes de escribir.

The circle painting's colors and sizes ____, which is why it differs from a ____ — you cannot ____ what the next circle will look like.

Di tu respuesta primero: Mi respuesta es ____ porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: The pattern unit is 3 shapes long, and $12 \div 3 = 4$ with no remainder, so shape 12 ends a complete unit — it is the blue triangle.

Evidence: A strong answer divides 12 by 3, gets remainder 0, and explains that means the LAST shape in the unit. A weak answer just guesses the shape without dividing.

Reasoning: This evidence connects the definition of repetition to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- The circle painting's colors and sizes ____, which is why it differs from a ____ — you cannot ____ what the next circle will look like.
- My answer is ____.

Mi respuesta es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- My claim is ____.
- I know because ____.

Mi afirmación es ____.

Lo sé porque ____.

- So ____.

Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
Mi afirmación es ____.
 - **My math evidence is ____.**
Mi evidencia matemática es ____.
 - **This proves ____ because ____.**
Esto demuestra ____ porque ____.
-
-
-

4. Check Your Explanation

- ☐ I answered the question.
Respondí la pregunta.
- ☐ I used math evidence: numbers, an equation, a model, or a comparison.
Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- ☐ I explained how the evidence proves my answer.
Explicué cómo la evidencia demuestra mi respuesta.
- ☐ I used at least two math words.
Usé por lo menos dos palabras matemáticas.
- ☐ I reread complete sentences and punctuation.
Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** repetition
2. **Notes 2:** pattern
3. **Notes 3:** pattern unit
4. **Notes 4:** rhythm
5. **Notes 5:** predictability
6. **Watch for this mistake:** *Calling every repeated design a 'pattern.' One identical shape repeated is repetition; a pattern needs a unit of shapes repeating in a regular order; and rhythm varies the shapes' size and order, so it is neither.*
7. **Try It 1:**
8. **Try It 2:**